



VNUHCM - UNIVERSITY OF SCIENCE
MID-TERM EXAMINATION
Term : 3 - Academic year: 2024-2025

MÃ LƯU TRỮ
(do phòng KT-DBCL ghi)

Course: Applied Statistics for Engineers and Scientists II

Course ID: STAT452

Time: 60 minutes

Date: 18/06/2025

Note: *Students are* [☐ **allowed** / ☒ **not allowed**] *to use course materials.*

Student name: Student ID:

Problem 01. (4pts) Consider the following dataset of self-study time x_1 , tutoring time x_2 , and math test score y for the following 5 students:

Student	x_1	x_2	y
1	5	1	5.0
2	9	0	6.0
3	7	7	9.5
4	9	7	10
5	3	8	7.5

(a) (2pts) Consider the following regression model:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Find the estimated values $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$. From there, calculate the R^2 coefficient of the model.

(b) (1pt) Now, consider the following simple linear regression model:

$$y = \beta_0 + \beta_1 x_1 + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$$

When comparing the two models, use the results from the `anova` function in R below to conclude which model is more appropriate:

Model	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1: <code>y ~ x1</code>	3	16.7941				
2: <code>y ~ x1 + x2</code>	2	0.2813	1	16.513	117.42	0.008409

(c) (1pt) Predict the math test score for a student who studies on their own for 5 hours and attends tutoring for 6 hours.

Problem 02. (3pts) The population of a city over 4 years is given in the table below:

Year (x)	Population y (unit: million people)
1	2.98
2	4.65
3	5.0
4	6.5

(a) (2pts) Consider the following exponential model:

$$y = \exp(\beta_0 + \beta_1 x + \epsilon), \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Convert this model to a linear model and find the estimated values $\hat{\beta}_0, \hat{\beta}_1$.

- (b) (1pt) Based on the estimated values, predict the population of this city in the 5th year.

Problem 03. (2pts)

- (a) (1pt) Consider a dataset with 4 columns x_1, x_2, x_3 and y , where $y \in \{0, 1\}$ is a binary response variable. Using the functions `glm` and `summary` in R, we obtain the following output:

Call:

```
glm(formula = y ~ x1 + x2 + x3, family = "binomial", data = data)
```

Deviance Residuals:

```

      Min       1Q   Median       3Q      Max
-1.5802  -0.8848  -0.6382   1.1575   2.1732

```

Coefficient Estimates and Significance:

```

              Estimate Std. Error z value Pr(>|z|) Signif.
(Intercept) -3.449548   1.132846  -3.045   0.00233    **
x1           0.002294   0.001092   2.101   0.03564     *
x2           0.777014   0.327484   2.373   0.01766     *
x3          -0.560031   0.127137  -4.405  1.06e-05    ***

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Write down the estimated logistic regression model from the data and interpret the estimated coefficients.

- (b) (1pt) Predict whether the following observations belong to class 0 or class 1:

	x_1	x_2	x_3
1	640	3.35	3
2	600	3.62	3
3	700	3.56	1
4	620	3.17	2

Problem 04. (1pt) Consider the linear regression model:

$$y = \beta_0 + \beta_1 x + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$$

For some $\lambda > 0$, we have the corresponding Ridge loss function:

$$L = \sum_{k=1}^n (y_k - \beta_0 - \beta_1 x_k)^2 + \lambda(\beta_0^2 + \beta_1^2)$$

Show that the estimated values of β_0, β_1 are:

$$\begin{aligned}
 \hat{\beta}_0 &= \frac{n}{n + \lambda} (\bar{y} - \hat{\beta}_1 \bar{x}) \\
 \hat{\beta}_1 &= \frac{S_{xy} + \frac{n\lambda}{n+\lambda} \bar{x}\bar{y}}{S_{xx} + \lambda \left[\frac{n}{n+\lambda} (\bar{x})^2 + 1 \right]} \\
 &==== \text{END} ====
 \end{aligned}$$